



Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering
2024-2025

Experiment No. 2
Implement Newton's Method
Date of Performance:
Date of Submission:



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Aim: To implement Newton's Method for optimization.

Theory:

Newton's method is an iterative optimization technique that uses the first and second derivatives (Hessian matrix) of a function to find its minima or maxima.

- Algorithm Steps:

1. Start with an initial guess for parameters, x_k .
2. Update parameters using:

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k).$$

where $\nabla^2 f(x_k)$ is the Hessian matrix of second derivatives.

- Advantages: Rapid convergence for well-conditioned problems.

- Challenges: Computationally expensive for large-dimensional problems due to the inversion of the Hessian matrix.

Procedure/Algorithm:

1. Define the function to be minimized and its second derivative (Hessian).
2. Initialize parameters.
3. Repeat until convergence:
 - Compute the gradient and Hessian.
 - Update parameters using the Newton update rule.
4. Display the optimized parameters and the value of the function.

Code:

Function: $f(x)=x^3-6x^2+4x+12$

Gradient (1st derivative): $f'(x)=3x^2-12x+4$

Hessian (2nd derivative): $f''(x)=6x-12$

```
def function(x):
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```
    return x**3 - 6*x**2 + 4*x + 12
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```
def gradient(x):
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```
    return 3*x**2 - 12*x + 4
```



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```
def hessian(x):
    return 6*x - 12

x = float(input("Enter the initial value of x: "))
iterations = int(input("Enter the number of iterations: "))

for i in range(iterations):
    grad = gradient(x)
    hess = hessian(x)

    if hess == 0: # Avoid division by zero
        print("Hessian is zero, Newton's Method cannot proceed.")
        break

    x_new = x - grad / hess

    if i < 10 or i == iterations - 1:
        print(f"Iteration {i+1}: x = {x_new:.6f}, f(x) = {function(x_new):.6f}")

    if abs(x_new - x) < 1e-6:
        print("Converged to optimal solution.")
        break

    x = x_new

print(f"\nFinal optimized x = {x:.6f}")
print(f"Function value at optimized x: f({x:.6f}) = {function(x):.6f}")
```

Output:

```
Enter the initial value of x: 1
Enter the number of iterations: 4
Iteration 1: x = 0.166667, f(x) = 12.504630
Iteration 2: x = 0.356061, f(x) = 12.708709
Iteration 3: x = 0.366970, f(x) = 12.709297
Iteration 4: x = 0.367007, f(x) = 12.709297

Final optimized x = 0.367007
Function value at optimized x: f (0.367007) = 12.709297
```



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Conclusion: Newton's Method quickly finds the minimum using derivatives, but it may fail if the second derivative is zero or the starting point is bad

Answer the following questions:

1. What is the advantage of using the Hessian in Newton's Method?

The Hessian helps Newton's Method adjust the step size dynamically, leading to faster and more accurate convergence compared to methods that use only the gradient. Compared to Gradient Descent, which uses only the first derivative (gradient) and takes fixed small steps, Newton's Method uses the second derivative (Hessian) to adjust step size dynamically. This makes Newton's Method faster and more precise, especially near the optimal point, while Gradient Descent may take more iterations or get stuck in slow convergence. However, Newton's Method requires computing the Hessian, which can be complex for high-dimensional problems.

2. What is the limitation of Newton's Method?

Newton's Method is powerful but has some limitations. It requires calculating the Hessian, which can be slow and difficult for complex problems. If the Hessian is zero or very small, the method fails because it cannot update the value properly. Also, if the starting point is far from the actual minimum, Newton's Method may not work well and could move in the wrong direction instead of finding the best solution.